

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

Practical – VIII

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Class:-TYIT	Batch:-1
Date of Assignment:-30-06-26	Date/Time of Submission:-30-06-26

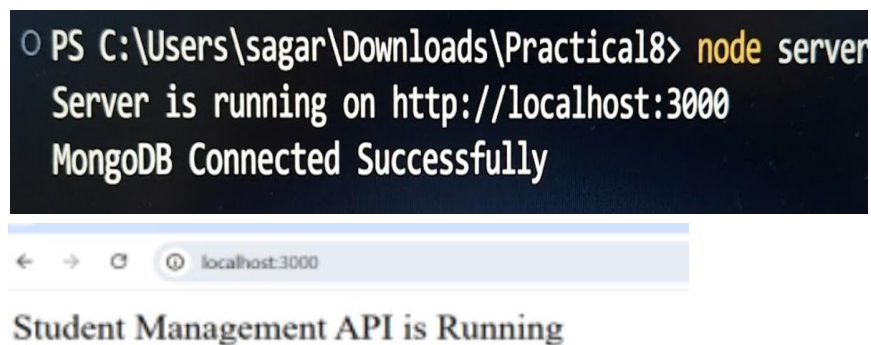
8.API Creation and Integration

a. Create a simple REST API using Express

Code:-

```
1  const express = require("express");
2  const mongoose = require("mongoose");
3  const cors = require("cors");
4  const app = express();
5  const PORT = 3000;
6  // Middleware
7  app.use(cors());
8  app.use(express.json());
9  // Connect MongoDB
10 mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB") .then(() => {
11   console.log("MongoDB Connected Successfully"); })
12   .catch((error) => {
13     console.log("MongoDB Connection Failed");
14     console.log(error);
15   });
16 // Simple REST API
17 app.get("/", (req, res) => {
18   res.send("Student Management API is Running"); });
19 app.listen(PORT, () => {
20   console.log(`Server is running on http://localhost:${PORT}`); });
21
```

Output:-



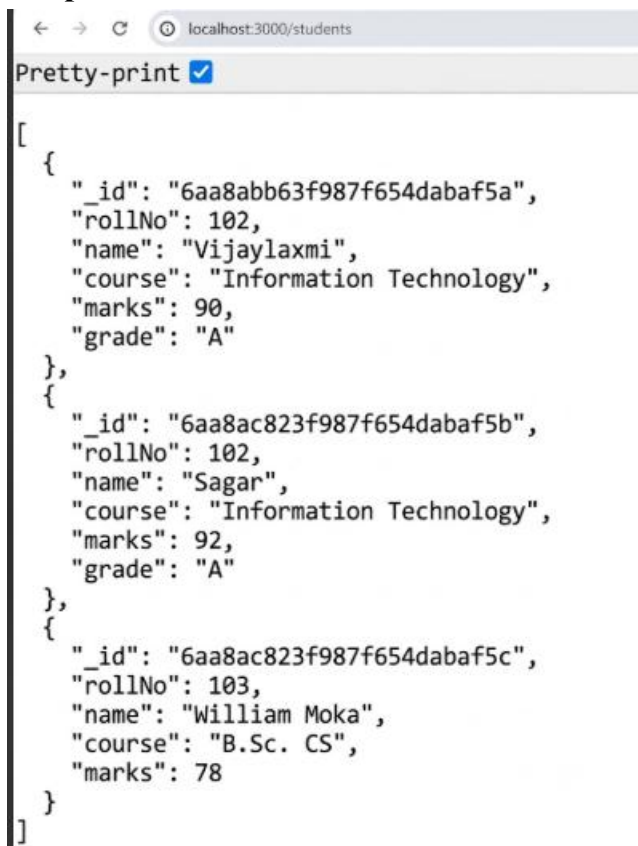
The image shows the output of the code. The top part is a terminal window with the command `node server` and the output: `Server is running on http://localhost:3000` and `MongoDB Connected Successfully`. The bottom part is a browser window showing the URL `localhost:3000` and the message `Student Management API is Running`.

b. Implement a GET API to retrieve data.

Code:-

```
1 app.get("/students", async (req, res) => {  
2   try {  
3     const students = await mongoose.connection.db  
4       .collection("students")  
5       .find()  
6       .toArray();  
7     res.json(students);  
8   }  
9   catch (error) {  
10    res.status(500).json({  
11      message: "Error fetching students"  
12    });  
13  }  
14 });  
15
```

Output:-



localhost:3000/students

Pretty-print ☒

```
[  
  {  
    "_id": "6aa8abb63f987f654dabaf5a",  
    "rollNo": 102,  
    "name": "Vijaylaxmi",  
    "course": "Information Technology",  
    "marks": 90,  
    "grade": "A"  
  },  
  {  
    "_id": "6aa8ac823f987f654dabaf5b",  
    "rollNo": 102,  
    "name": "Sagar",  
    "course": "Information Technology",  
    "marks": 92,  
    "grade": "A"  
  },  
  {  
    "_id": "6aa8ac823f987f654dabaf5c",  
    "rollNo": 103,  
    "name": "William Moka",  
    "course": "B.Sc. CS",  
    "marks": 78  
  }  
]
```

c. Fetch data from the API in a React application.

Code:-

```
import { useEffect, useState } from "react";  
import axios from "axios";
```

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```
function App() {
  const [students, setStudents] = useState([]);
  useEffect(() => { axios.get("http://localhost:3000/students")
    .then((response) => {setStudents(response.data);})
    .catch((error) => {console.log(error);}); }, []);
  return (
    <div>
      <h1>Student List</h1>
      <table border="1">
        <thead>
          <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Course</th>
            <th>Marks</th>
          </tr>
        </thead>
        <tbody>
          {students.map((student) => (
            <tr key={student._id}>
              <td>{student.rollNo}</td>
              <td>{student.name}</td>
              <td>{student.course}</td>
              <td>{student.marks}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}
export default App;
```

Output:-



Student List

Roll No	Name	Course	Marks
102	Vijaylaxmi	Information Technology	90
103	William Moka	B.Sc. CS	78
104	Sagar	B.Sc. IT	85

d. Send data from React to the server using a POST request.

Code:-

1) Replace App.jsx code

```
import { useEffect, useState } from "react";
import axios from "axios";

function App() {
  const [students, setStudents] = useState([]);
  const [rollNo, setRollNo] = useState("");
  const [name, setName] = useState("");
  const [course, setCourse] = useState("");
  const [marks, setMarks] = useState("");

  // Get students
  const getStudents = () => {
    axios.get("http://localhost:3000/students")
      .then((response) => {setStudents(response.data);})
      .catch((error) => {console.log(error);});
  };

  useEffect(() => {getStudents();}, []);

  // Add student
  const addStudent = () => {
    const newStudent = {
      rollNo: Number(rollNo),
      name: name,
      course: course,
      marks: Number(marks)
    };
  };
}
```

```
    axios.post("http://localhost:3000/students", newStudent)
    .then((response) => {alert(response.data.message); setRollNo("");
    setName("");
    setCourse("");
    setMarks("");
    getStudents();
    })
    .catch((error) => {
    console.log(error);
    });
    };
    return (
    <div>
    <h1>Student Management System</h1>
    <h2>Add Student</h2>
    <input
    type="number"
    placeholder="Roll No"
    value={rollNo}
    onChange={ (e) => setRollNo(e.target.value)}    />
    <br /><br />
    <input
    type="text"
    placeholder="Student Name"
    value={name}
    onChange={ (e) => setName(e.target.value)}    />
    <br /><br />
    <input
    type="text"
    placeholder="Course"
    value={course}
    onChange={ (e) => setCourse(e.target.value)}    />
    <br /><br />
    <input
    type="number"
    placeholder="Marks"
    value={marks}
    onChange={ (e) => setMarks(e.target.value)}    />
    <br /><br />
    <button onClick={addStudent}>
```

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```
Add Student
</button>
<hr />
<h2>Student List</h2>
<table border="1">
  <thead>
    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Course</th>
      <th>Marks</th>
    </tr>
  </thead>
  <tbody>
    {students.map((student) => (
      <tr key={student._id}>
        <td>{student.rollNo}</td> <td>{student.name}</td>
        <td>{student.course}</td> <td>{student.marks}</td>
      </tr>
    ))}
  </tbody>
</table>
</div>
);
}
export default App;
```

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2) Add post method in server.js file above app.listen()

```
server.js > ...  
app.post("/students", async (req, res) => {  
  try {  
    const student = req.body;  
    await mongoose.connection.db  
      .collection("students")  
      .insertOne(student);  
    res.json({  
      message: "Student added successfully",  
      student: student  
    });  
  }  
  catch (error) {  
    res.status(500).json({  
      message: "Error adding student"  
    });  
  }  
});
```

Output:-

Student Management System

Add Student

Roll No

Student Name

Course

Marks

Student List

Roll No	Name	Course	Marks
102	Sagar	Information Technology	90
103	William Moka	B.Sc. CS	78
104	Soham	Biotechnology	95
105	Aisha Khan	Civil Engineering	88